TERMS OF REFERENCE FOR SOCIAL IMPACT ASSESSMENT

COURSE NAME:

ENVIRONMENTAL AND SOCIAL IMPACT ASSESSMENT

COURSE CODE: HS-11

SUBMITTED BY:

GROUP 6

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Terms of Reference (ToR) for Social Impact Assessment (SIA) of a Tidal Energy Project in Andaman and Nicobar Islands, India

1. Objectives

The ESIA aims to:

1. Ensure compliance with India's environmental regulations and IFC Performance Standards, especially Performance Standard 7 (Indigenous Peoples).
2. Identify, assess, and propose measures to mitigate environmental and social risks, including impacts on biodiversity and Indigenous Peoples.
3. Develop an Environmental Management Plan (EMP) and a culturally appropriate engagement strategy for Indigenous Peoples.
4. Ensure the project contributes to the local economy and climate change mitigation objectives.

2. General Information

- **Project Developer**: Sun Energy Systems
- **Location**: Chouldari, Sri Vijaya Puram, Andaman and Nicobar Islands.
- **Project Type**: Renewable Energy (Tidal)
- **Funding**: International Monetary Fund
- **Energy Output**: Estimated at **5-10 MW**.
- **Estimated Cost of ESIA**: **2,11,09,070 INR** (includes *surveys, community consultations, and additional studies*).

3. Introduction

The Andaman and Nicobar Islands host ***ecologically sensitive regions*** and ***Indigenous Peoples*** with distinct cultural identities. This ESIA is designed to evaluate environmental and social risks in alignment with **IFC PS7**, ensuring the project avoids adverse impacts on vulnerable groups and habitats.

4. Project Description

- **Technology**: Horizontal axis tidal turbines, with floating or seabed-mounted installations.
- **Capacity**: Approx. **5 MW initially**, scalable to **10 MW**.
- **Components**: Turbines, underwater cables, an onshore substation, and transmission lines.
- **Site Location**: **Chouldari,** chosen to minimize impact on coral reefs and sensitive habitats.
- **Construction**: Expected duration of **24 months**, employing **100 workers**, with **30% local hiring** commitment.
- **Operational Life**: Estimated at **25 years**.
- **Project Phases*:***

  i. Development
  ii. Construction
  iii. Operation
  iv. Decommissioning.

5. Analysis of Alternatives

The study will evaluate:

1. **Alternative Locations**: Comparing ecological sensitivity, tidal current intensity, and social impacts.
2. **Alternative Designs**: Floating vs. seabed-mounted turbines.
3. **"No Project" Scenario**: Assessing potential loss of renewable energy and continued reliance on diesel generators in the region.

6. Description of Environment

- **Land**: Coastal erosion risk, vegetation clearance limited to **<5 hectares**.
- **Air**: Baseline PM 2.5 levels under **30 µg/m³**, with temporary construction emissions.
- **Water**: Marine biodiversity (coral, seagrass), Tidal Current data collected over **12 months**.
- **Noise**: Marine noise baseline at **ambient 60 dB**, Turbine noise at operational levels **80-100 dB underwater**, assessed for marine life impact.

7. Anticipated Environmental Impacts and Mitigation Measures

- **Impacts**
  - *i.* **Marine Biodiversity**: Risk to fish and marine mammals due to turbine operation.
  - *ii.* **Coral Reefs**: Risk of sediment displacement during construction.
  - *iii.* **Indigenous Communities**: Possible disruption to cultural practices, fishing zones.
- **Mitigation Measures**
  - *i.* Marine life exclusion zones around turbines during construction.
  - *ii.* Construction windows avoiding breeding/spawning seasons.
  - *iii.* Capacity-building programs for Indigenous Peoples to participate in project benefits.

8. Environmental Monitoring Program

- **Cost**: **$50,000** i.e. **42,20,250 INR** annually.
- **Institutional Arrangements**: Roles and responsibilities of the developer and relevant agencies.
- **Capacity Building and Training**: For staff and local stakeholders on environmental management.
- **Budget and Timeline**: Estimated costs and schedule for EMP implementation.
- **Parameters**:
  - *i.* Marine life monitoring using underwater cameras.
  - *ii.* Bi-annual water quality testing for turbidity, salinity, and dissolved oxygen.
  - *iii.* Noise monitoring during construction and operation.

9. Additional Studies

1. **Biodiversity Impact Study**: **$75,000** i.e. **63,30,375 INR** to map sensitive coral and marine habitats.
2. **Indigenous Peoples' Livelihood Assessment**: **$50,000** i.e. **42,20,250 INR** documenting cultural sites, fishing practices, and subsistence resources.
3. **Gender Assessment**: **$25,000** i.e. **21,10,125 INR** ensuring inclusivity in benefit-sharing.

10. Project Benefits

- **Energy Access**: Reducing dependence on diesel by **30%**.
- **Employment**: 50 local jobs during construction and 20 during operation, with training programs for skill development.
- **Environmental**: Preventing **10,000 tons of CO_2 emissions annually**.

11. Environmental Management Plan (EMP)

Components

1. **Waste Management**: Zero marine dumping, onshore processing of construction debris.
2. **Biodiversity Protection**: Continuous coral monitoring and turbine design improvements based on marine studies.
3. **Community Engagement**: Establishing grievance redress mechanisms, **$20,000** i.e. **16,88,100 INR** annually for consultation programs.

12. Summary and Conclusions

The ESIA will summarize key impacts, mitigation strategies, and project benefits. It will emphasize compliance with IFC PS7 to respect Indigenous Peoples' rights and cultural heritage.

13. Disclosure of Consultants

- **Team Composition**:
  - *i.* Marine Biologist
  - *ii.* Social Anthropologist, expert on Indigenous Peoples in the Andaman Islands.
  - *iii.* Environmental Engineer
  - *iv.* Cost Analyst
- **Budget Allocation**: Consultant fees set at **$120,000** i.e. **1,01,28,600 INR**

This Terms of References has been prepared from drawing parallels from ***The MeyGen Tidal Energy Project (Scotland, UK)*** as no alternate projects could be found with respect to the area selected. Adjustments according to the site and resources have been made with the IFC Performance Standard 7 and our own understanding.

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